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CSE3010 – COMPUTERVISION

Vityarthi PROJECT REPORT

Dual-Domain Digital Image Forgery Detection Engine

Using Error Level Analysis (ELA) and SIFT–RANSAC Feature Geometry

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1 Introduction

Digital images can be manipulated using conventional editing software and modern generative tools. Such manipulation can be difficult to identify through visual inspection alone, particularly when a region is copied within the same image or when foreign content is inserted and the image is subsequently compressed.

This project develops a deterministic, dual-domain image-forensics engine for detecting two important forms of manipulation: **copy-move forgery** and **splicing/resaving artifacts**. The system combines Error Level Analysis (ELA), which examines differences produced by JPEG recompression, with SIFT feature matching and RANSAC-based geometric verification.

The approach was selected because its main decisions are mathematically interpretable and directly demonstrate computer-vision concepts such as image preprocessing, feature extraction, descriptor matching, geometric transformation, and outlier rejection. The application operates as a modular command-line interface (CLI), making the complete pipeline reproducible on commodity CPU hardware.

2 Problem Statement and Objectives

Manual inspection of digital images is unreliable for detecting subtle tampering. Two difficult cases are localized compression inconsistencies and duplicated regions that have been translated, rotated, or otherwise transformed.

The project addresses these problems through the following objectives:

- Validate and preprocess input images before forensic analysis.
- Detect copy-move regions using SIFT keypoints and 128-dimensional descriptors.
- Filter candidate matches using KNN matching, Lowe's ratio test, spatial distance pruning, and RANSAC.
- Detect compression-level discrepancies using ELA.
- Produce a visual forensic report containing the original image and diagnostic overlays.
- Provide deterministic execution, modular implementation, and automated validation tests.

3 Functional and Non-Functional Requirements

3.1 Functional Requirements

1. **Input and preprocessing:** Validate file availability, enforce a minimum image size of 128×128 pixels, convert between BGR/grayscale/RGB representations, and handle constrained resizing.
2. **SIFT-RANSAC copy-move detection:** Extract SIFT features, perform KNN matching with $k = 3$, apply Lowe's ratio test, remove nearby self-similar matches, and verify geometric consistency with RANSAC homography estimation.
3. **ELA residual analysis:** Recompress an image at a selected JPEG quality, calculate absolute pixel differences, amplify the residual map, and calculate a mean discrepancy score.
4. **Visualization:** Generate a side-by-side diagnostic canvas with the original image, forensic overlays, and analysis information.

3.2 Non-Functional Requirements

- **Performance:** Target analysis time below 2.5 seconds for a standard 1024×768 image on commodity CPU hardware.
- **Reliability:** Produce deterministic and reproducible metrics across repeated runs.
- **Maintainability:** Separate preprocessing, ELA, copy-move, CLI, and visualization responsibilities.
- **Error handling:** Gracefully handle invalid paths, corrupted input, zero keypoints, and invalid parameters.

4 System Architecture and Workflow

The system is organized around a CLI dispatcher. The dispatcher loads and validates the image, invokes one or both forensic modules, and sends their results to the visualization component.

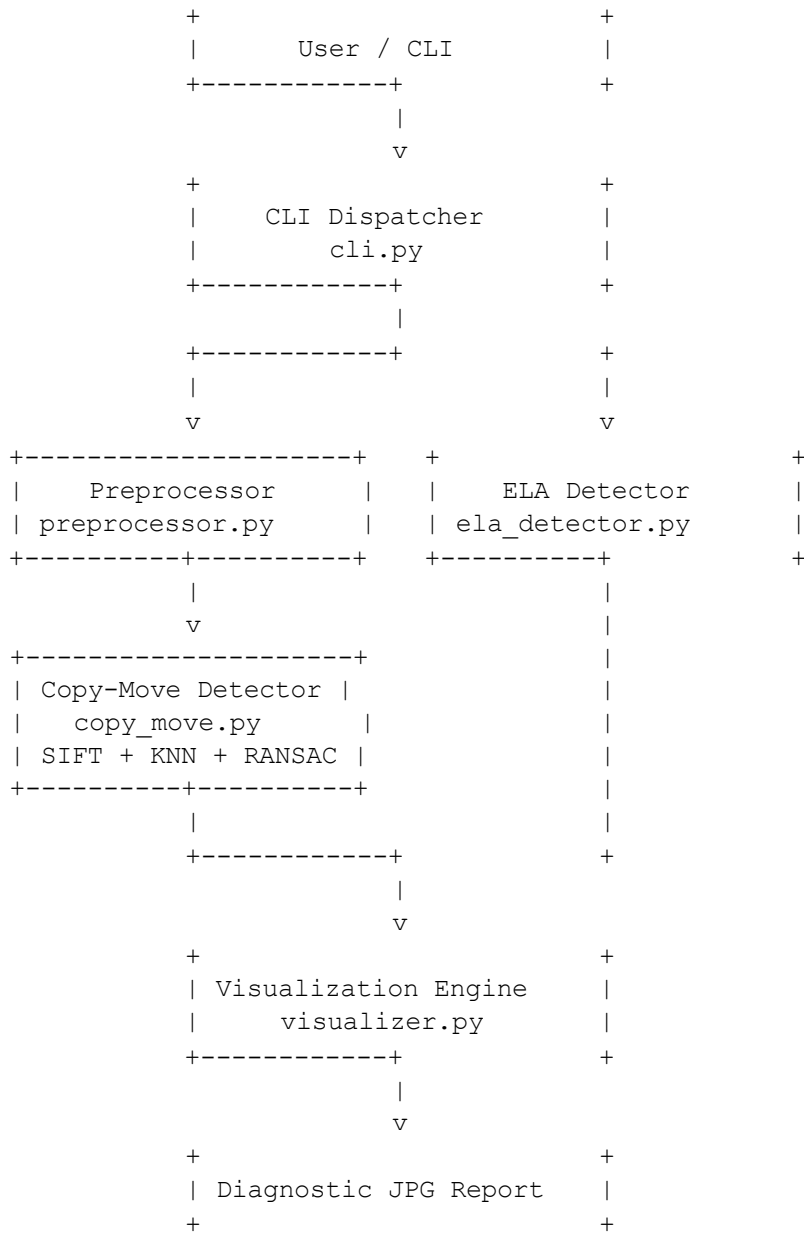


Figure 1: System architecture.

4.1 Process Workflow

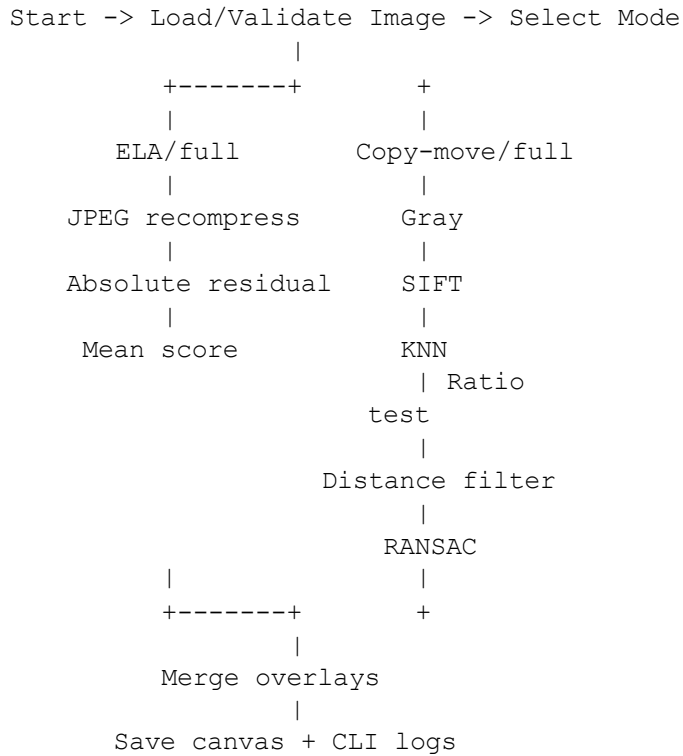
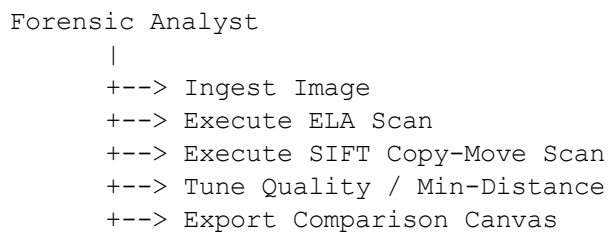


Figure 2: Forensic processing workflow.

5 Design Diagrams

5.1 Use Case and Component Design

The primary actor is a forensic analyst who supplies an image, selects the analysis mode, optionally tunes parameters such as JPEG quality or minimum matching distance, and exports the diagnostic result.



Components:

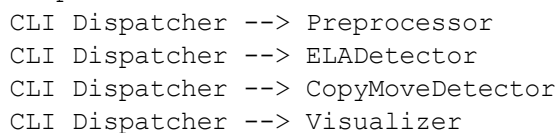


Figure 3: Simplified use-case and component relationships.

5.2 Sequence of Execution

```

User -> CLI: python cli.py -i input.jpg -m full
CLI -> Prep: load_image(input.jpg)
Prep -> CLI: return image array
CLI -> ELA: compute_ela(input.jpg, quality=90)
ELA -> CLI: return residual canvas and score CLI
-> CM: detect_copy_move(image, min_dist=30)
CM -> CLI: return annotated canvas and inlier count
CLI -> Vis: save_comparison(original, result, report.jpg) Vis
-> CLI: file written
CLI -> User: verdicts and output path
  
```

6 Algorithms and Implementation

The implementation uses Python 3.12 with OpenCV 4.8+, NumPy, and Pillow. The project is divided into modules such as `preprocessor.py`, `ela_detector.py`, `copy_move.py`, `visualizer.py`, and `cli.py`.

6.1 Error Level Analysis

JPEG compression divides an image into 8×8 blocks and applies the Discrete Cosine Transform (DCT). Quantization of frequency coefficients means that recompressing an image can produce pixel-level differences. If an inserted region has a different compression history, its residual pattern may differ from surrounding content.

The residual is calculated as

$$\Delta(x, y) = |I_{\text{orig}}(x, y) - I_{\text{recompress}}(x, y, Q)|.$$

For visualization, the residual is amplified and limited to the valid 8-bit intensity range:

$$\Delta_{\text{scaled}}(x, y) = \min(255, \Delta(x, y) \times \frac{255}{\max(\Delta)}).$$

The mean discrepancy score is used as a quantitative diagnostic measure. The $\overline{\text{ELA}}$ result is interpreted together with the image context rather than as a standalone proof of manipulation.

6.2 SIFT Feature Extraction

The copy-move module converts the image to grayscale and extracts scale-space keypoints and 128-dimensional SIFT descriptors. SIFT is used because its descriptors are designed to remain useful under changes in scale and illumination.

6.3 KNN Matching and Lowe's Ratio Test

Feature descriptors are compared using K-nearest-neighbor matching. A value of $k = 3$ is used so that the identity self-match can be bypassed when descriptors are compared within the same image.

Candidate matches are filtered according to Lowe's ratio criterion:

$$\frac{D(f_i, f_{m2})}{D(f_i, f_{m1})} < 0.75,$$

where f_{m1} and f_{m2} denote the nearest and second-nearest candidate descriptors.

6.4 Spatial Filtering and RANSAC

A minimum spatial separation of approximately 30 pixels is applied because adjacent keypoints commonly belong to the same physical region. Remaining point correspondences are evaluated using homography estimation with RANSAC:

$$p \sim_2 = H p \sim_1.$$

Correspondences inconsistent with the estimated transformation within a 5-pixel error margin are rejected as outliers. The remaining inliers provide geometric evidence for a duplicated region.

7 Results and Evaluation

A recorded execution was performed on a 600×400 test image containing two synthetic identical circles with the text "TEST." The CLI output was:

```
CSE3010 Image Forgery Detection CLI
Input: samples/test.jpg
Resolution: 600x400 | Channels: 3
ELA discrepancy score: 0.30
ELA verdict: CLEAN
Detected inlier matches: 52
Copy-move verdict: FORGERY DETECTED
Inspection report saved to: outputs/full_report.jpg
```

The copy-move detector found **52 RANSAC inliers** connecting the duplicated regions. This indicates that the detected feature correspondences were geometrically consistent with a common transformation.

The ELA score was **0.30**, producing the recorded `CLEAN` verdict. The supplied test description explains that the synthetic image was generated on a uniform digital canvas, resulting in low compression-residual variance. This illustrates why the two

forensic signals are complementary: copy-move analysis targets repeated feature geometry, while ELA examines recompression discrepancies.

7.1 Testing

Four automated tests were recorded as passed:

Test	Purpose	Result
test_preprocessor_invalid_path	Missing-image error handling	Passed
test_grayscale_conversion	Validate 3-channel to 2D conversion	Passed
test_ela_computation	Validate score and output canvas	Passed
test_copy_move_synthetic_blank	Prevent crashes on blank frames	Passed
-	-	-

Table 1: Unit-test summary.

Recorded test execution:

```
tests\test_pipeline.py [100%]  
4 passed in 0.57s
```

8 Challenges and Design Rationale

8.1 Self-Matching of SIFT Features

When descriptors from the same image are matched directly, a keypoint can select itself as the nearest neighbor with distance zero. Using $k = 3$ and evaluating distinct neighbors prevents this identity match from dominating the ratio test.

8.2 False Positives in Repetitive Textures

Repeated structures such as leaves, bricks, or symmetric patterns can generate visually plausible but incorrect feature correspondences. Spatial pruning reduces nearby duplicate matches, while RANSAC removes correspondences that do not share a consistent geometric transformation.

8.3 Classical CV versus Deep Learning

The project intentionally uses classical computer vision instead of an opaque learned classifier. The selected methods are deterministic, interpretable, locally executable, and demonstrate several core CSE3010 concepts without requiring GPU-based model training.

9 Learnings and Future Enhancements

The project provided practical experience with SIFT descriptors, Lowe’s ratio test, KNN matching, homography estimation, RANSAC outlier rejection, JPEG compression effects, and test-driven modular implementation. It also demonstrated how different forensic cues can produce different observations for the same image.

Planned enhancements include:

- **DCT frequency heatmaps:** Perform block-wise 8×8 DCT analysis to localize compression anomalies.
- **Deep feature embeddings:** Add an optional ResNet or ViT-based semantic tampering detector alongside the classical pipeline.
- **Batch processing:** Add recursive directory analysis and structured JSON/CSV forensic reports.

10 Conclusion

This project presents a compact, deterministic image-forensics pipeline combining ELA-based compression analysis with SIFT–RANSAC feature geometry. The implementation provides separate preprocessing, detection, and visualization modules and includes automated tests for important edge cases. The recorded synthetic evaluation successfully detected copy-move duplication through 52 geometrically consistent inlier matches while producing a low ELA discrepancy score for the uniformly generated test image. The project therefore demonstrates how complementary classical computer-vision techniques can be combined into an interpretable forensic analysis tool.